

Biocompatibility of micro- and nano-particles in the colon. Part II.



Pathological colonic tissues were investigated with an Environmental Scanning Electron Microscope technique to verify the presence of inorganic, non-biodegradable pollutants, i.e. micro- and nano-debris of exogenous origin, after debris in liver and kidney had been discovered. In all, 18 samples of colon tissues affected by cancer and Crohn's disease were evaluated and found in all the cases to contain micro- and nano-particles. Their chemistry, detected with an X-ray microprobe, indicated a heterogeneous nature, whereas the size of the particles was homogeneous. Three control samples of healthy, young, cadavers were analysed and showed the absence of debris within the normal, healthy colon mucosa. The study reveals the presence of particulate debris, generally considered as biocompatible, in pathological specimens of human colon. The findings suggest a possible link between the presence of such particles and the underlying pathology in the cases analysed.